NUMERICAL ANALYSIS
PROBLEM SET 1

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1. The value of π is given by the alternating series:

$$\pi = 4\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} = 4\left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots\right)$$

(a) Use this series to construct an algorithm to approximate the value of π to within a specified tolerance, ε. The algorithm should be general to allow for any ε. Provide a theoretical upperbound ϵ on the error of your approximation. Hint. It may be useful to recall known properties of the partial sums of alternating series.

(b) Run your algorithm with a tolerance value of $\varepsilon = 5 \times 10^{-3}$ and show that your result is within ε of $\pi = 3.1415926\ldots$

(c) Make a plot of ε versus the number N of reuired terms in your algorithm on a log-log scale. What is the slope of the line that the resulting plot approaches a N increases? Plot the actual error as a function of N on the same axes, you may treat the stored value of π as "exact." How does the actual error compare to the estimated threshold?

Solution. Problem 1 □

2. Numerically determine which of the following limits approaches 1 faster. Then confirm the numerical evidence by determining the rate of convergences of each sequence

$$\lim_{n\to 0} \frac{\sin(x^2)}{x^2} \qquad \text{versus} \qquad \lim_{n\to 0} \frac{(\sin x)^2}{x^2}.$$

Solution. Problem 2 □

3. (a) Consider the function $f_c(x) = x^2 - 12x + c$. What are the zeros of f_c with $c = 35$? What are the zeros of f_c with the slight constant shift $c = 35 - 10^{-2}$? Relative to the size of change in the constant term c, how big is the change in the zeros of the function?

(b) consider the function $g_c(x) = x^2 - 10x + c$. What are the zeros of g_c with $c = 25$? What are the zeros of g_c with the slightest constant shift $c = 25 - 10^{-2}$? Relative to the size of change in the constant term, how big is the change in the zeros of the function?

(c) Compare the nature of the roots of f and g. How might this contribute to the relative changes in the zeros for each function?

Solution. Problem 3 □

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4. Identify the potential roundoff error problems in the following algorithm for calculating the roots of the quadratic equation $ax^2 + bx + c = 0$.

$$\begin{gathered}
\texttt{GIVEN:}\ \text{real coefficients } a, b, c \\
\texttt{STEP 1:}\ \text{calculate } disc = \sqrt{b^2 - 4ac} \\
\texttt{STEP 2:}\ \text{calculate } root1 = -2c/(b + disc) \\
\texttt{STEP 3:}\ \text{calculate } root2 = (-b/a) - root1 \\
\texttt{OUTPUT:}\ root1 \text{ and } root2
\end{gathered}$$

Note that this algorithm uses the fact that the sum of the roots of $ax^2 + bx + c = 0$ is equal to $-b/a$.

Solution. Problem 4 □

5. (a) Plot the function $f(x) = \ln(1 + x) - \cos(x) - x + 1$ over the interval $-5 \times 10^{-6} \leq x \leq 5 \times 10^{6}$ using 1001 uniformly spaced points in x.

(b) Reformulate f to avoid the cancellation error, and then repeat part (a).

(c) Describe the issues with the original formulation in part (a) and how they were resolved with the reformulation in part (b).

Solution. Problem 5 □

6. (a) Determine the third-degree Taylor polynomial associated remainder term for the function $f(x) = \sqrt{(1 + x)}$. Use $x_0 = 0$.

(b) Using the result of part (a), approximate $\sqrt{1.5}$, and compute the theoretical error bound associated with this approximation. Compare the theoretical error bound with the actual error.

(c) Compute the following limit and determine the corresponding rate of convergence:

$$\lim_{x \to 0} \frac{\sqrt{1 + x} - 1 - \frac{1}{2}x}{x^2}$$

Solution. Problem 6 □

7. Consider the linear system $Ax = b$ with $A = \begin{bmatrix} 3.4 & 2.8 \\ 8.0 & 6.6 \end{bmatrix}$, $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, and $b = \begin{bmatrix} 3.4 \\ 8.0 \end{bmatrix}$.

(a) Determine the analytic solution by hand. (How do we know that it is unique, i.e. that the problem is well-posed?)

(b) Now solve the system with $b = \begin{bmatrix} 3.41 \\ 8.0 \end{bmatrix}$. You may use your computer.

(c) Given the difference in solutions for parts (a) and (b), comment on the conditioning of the problem. What is $\det[A]$ and for what values with the problem be ill-posed?

Solution. Problem 7 □